Development of an Ultra-Low-Cost Biodegradable Packaging Bag Using Jackfruit Waste and Coconut Shell Biochar

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Abstract

Plastic pollution caused by single-use polyethylene bags presents a severe environmental challenge, particularly in developing countries where cost constraints limit the adoption of eco-friendly alternatives. Existing biodegradable bags are often environmentally sound but economically impractical for mass adoption. This research explores the development of an ultra-low-cost biodegradable packaging bag derived from agricultural waste materials readily available in Bangladesh—specifically jackfruit string, jackfruit core, and jackfruit seed starch—combined with a functional biochar layer produced from coconut shells to enhance water resistance and thermal stability.

The proposed material system emphasizes simplicity, affordability, biodegradability, and local resource utilization. Preliminary formulation analysis suggests that the raw material cost per bag can be maintained at approximately 1 Bangladeshi Taka or lower, making it competitive with conventional plastic bags. This paper presents the material selection rationale, processing methodology, functional role of coconut shell biochar, and the environmental and economic significance of the approach. The study aims to serve as a foundation for further laboratory validation, mechanical testing, and collaborative development with academic researchers.

Keywords

Biodegradable bags, jackfruit waste, seed starch, coconut shell biochar, low-cost materials, sustainable packaging

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1. Introduction

Single-use plastic bags remain one of the most persistent contributors to environmental pollution due to their low degradation rate and widespread usage. Although biodegradable alternatives exist, most commercially available options rely on imported polymers, complex processing, or costly additives, limiting their scalability in low-income regions.

Bangladesh, an agrarian country with abundant agricultural waste, presents an opportunity for locally sourced biodegradable materials. Jackfruit (*Artocarpus heterophyllus*), the national fruit of Bangladesh,

produces significant quantities of waste in the form of strings, cores, and seeds. Meanwhile, coconut shells—often discarded or underutilized—are a rich source of carbon and can be converted into biochar.

This research proposes a biodegradable bag system that combines **jackfruit waste-based biopolymers** with a **coconut shell biochar coating layer** to achieve mechanical integrity, water resistance, and heat tolerance while maintaining extreme cost efficiency.

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2. Materials Selection and Rationale

2.1 Jackfruit Seed Starch

Jackfruit seeds contain a high starch content, which can function as a natural thermoplastic when gelatinized. Compared to corn or potato starch, jackfruit seed starch:

- is locally abundant
- requires no import cost
- utilizes waste material
- exhibits strong film-forming ability

The starch serves as the primary binding matrix of the bag.

2.2 Jackfruit String and Core Fibers

Jackfruit strings and cores are rich in cellulose fibers. When finely processed and incorporated into the starch matrix, these fibers:

- increase tensile strength
- reduce brittleness
- improve structural integrity
- lower overall starch consumption

Their inclusion transforms the material from a weak film into a usable bag structure.

2.3 Coconut Shell Biochar (Functional Layer)

A novel feature of this project is the use of **coconut shell-derived biochar** as a thin functional layer.

Coconut shell biochar is selected because:

- it has high carbon content
- it is hydrophobic compared to pure starch
- it provides thermal resistance
- it is extremely low-cost

The biochar is applied as a surface layer or composite blend to:

- reduce water absorption
- improve heat resistance
- slow down premature degradation during use

Importantly, biochar remains biodegradable and environmentally safe.

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3. Manufacturing Methodology (Proposed)

3.1 Starch Extraction

1. Jackfruit seeds are cleaned, dried, and ground.
2. Starch is extracted using water separation and sedimentation.
3. The starch is dried and stored.

3.2 Fiber Preparation

1. Jackfruit strings and cores are washed and sun-dried.
2. Fibers are mechanically shredded into fine particles.
3. Fibers are sieved to ensure uniform size.

3.3 Biochar Production

1. Coconut shells are carbonized under limited oxygen conditions.
2. The resulting biochar is finely powdered.

3. Particle size is controlled for uniform coating.

3.4 Bag Formation

1. Starch is gelatinized using heat and water.
2. Fibers are mixed into the starch matrix.
3. The composite is cast or pressed into thin sheets.
4. Biochar is applied as a coating or layered composite.
5. Sheets are dried and heat-pressed into bag form.

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4. Cost Analysis

A critical goal of this project is affordability.

Component	Cost Source	Estimated Cost per Bag
Jackfruit waste	Agricultural waste	Near zero
Seed starch	Local processing	Very low
Coconut shell biochar	Waste shells	Very low
Energy & water	Local methods	Minimal

Estimated total cost per bag: ≈ ≤ 1 BDT

This cost level directly addresses one of the main reasons biodegradable bags fail in developing markets.

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5. Environmental Impact

- Fully biodegradable under natural conditions
- Utilizes waste materials instead of food crops
- Reduces plastic pollution
- Biochar contributes to carbon sequestration
- Low energy manufacturing potential

The project aligns with circular economy principles.

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6. Limitations and Future Work

This study represents an early-stage material design concept. Future work will include:

- tensile strength testing
- water absorption analysis
- heat resistance evaluation
- biodegradation time studies
- optimization of biochar layer thickness

Collaboration with academic laboratories will be essential for advanced characterization.

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7. Significance for Developing Countries

The proposed biodegradable bag system is designed **not for luxury markets**, but for:

- street vendors
- rural markets
- developing economies

By prioritizing cost, simplicity, and local materials, this project addresses real-world adoption barriers.

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8. Conclusion

This research proposes a novel, ultra-low-cost biodegradable bag made from jackfruit agricultural waste and coconut shell biochar. The approach demonstrates that sustainability and affordability can coexist when local resources are intelligently utilized. With further scientific validation, this material system has the potential to serve as a scalable alternative to single-use plastic bags in resource-limited regions.

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